Group F

Power Query and Power BI Screenshots:

Members

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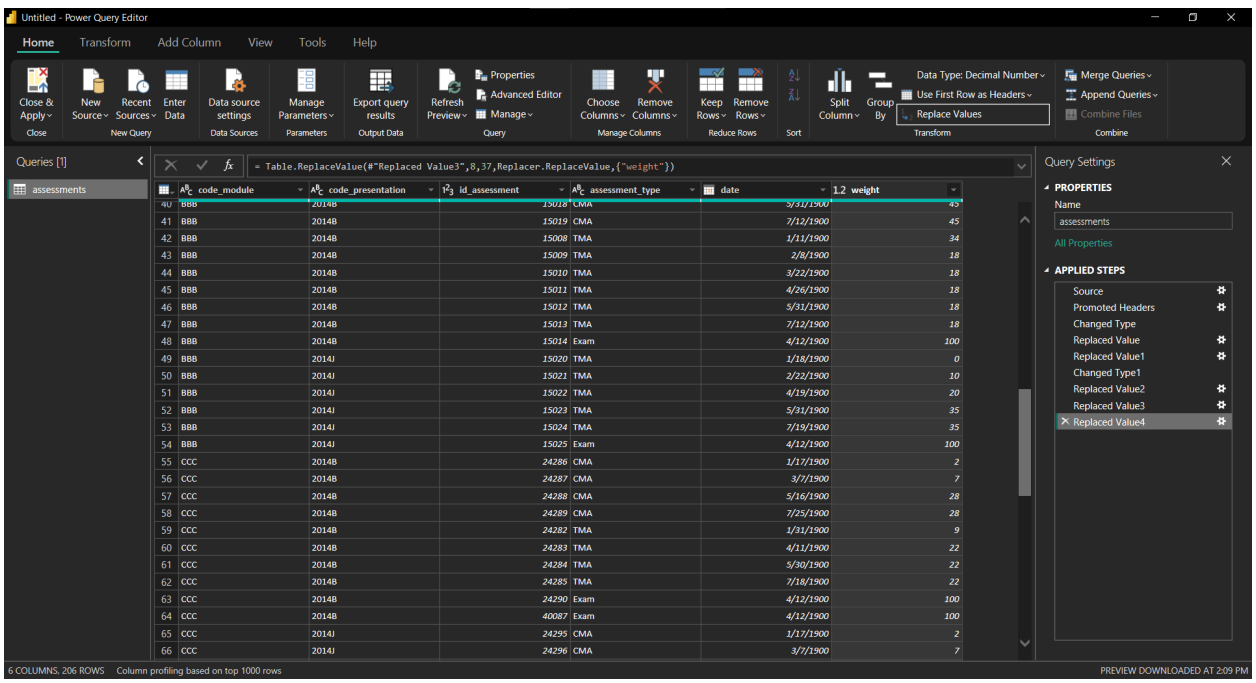
Hope Kimandi

Cleaning & Transformations

1. Replacing nulls



Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Queries [1]: assessments

```
= Table.ReplaceValue(#"Replaced Value3",8,37,Replacer.ReplaceValue,{"weight"})
```

	code_module	code_presentation	id_assessment	assessment_type	date	weight
40	BBB	2014B	15018	CMA	5/31/1900	45
41	BBB	2014B	15019	CMA	7/12/1900	45
42	BBB	2014B				34
43	BBB	2014B				18
44	BBB	2014B				18
45	BBB	2014B				18
46	BBB	2014B				18
47	BBB	2014B				18
48	BBB	2014B				100
49	BBB	2014J				0
50	BBB	2014J				10
51	BBB	2014J				20
52	BBB	2014J				35
53	BBB	2014J				35
54	BBB	2014J				100
55	CCC	2014B	24286	CMA	1/17/1900	2
56	CCC	2014B	24287	CMA	3/7/1900	7
57	CCC	2014B	24288	CMA	5/16/1900	28
58	CCC	2014B	24289	CMA	7/25/1900	28
59	CCC	2014B	24282	TMA	1/31/1900	9
60	CCC	2014B	24283	TMA	4/11/1900	22
61	CCC	2014B	24284	TMA	5/30/1900	22
62	CCC	2014B	24285	TMA	7/18/1900	22
63	CCC	2014B	24290	Exam	4/12/1900	100
64	CCC	2014B	40087	Exam	4/12/1900	100
65	CCC	2014J	24295	CMA	1/17/1900	2
66	CCC	2014J	24296	CMA	3/7/1900	7

Replace Values

Replace one value with another in the selected columns.

Value To Find: 0

Replace With: 46

OK | Cancel

Query Settings

PROPERTIES

Name: assessments

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Replaced Value1
- Changed Type1
- Replaced Value2
- Replaced Value3
- Replaced Value4

6 COLUMNS, 206 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:09 PM

Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Queries [3]: assessments, courses, studentAssessment

```
= Table.RemoveColumns(#"Changed Type",{"is_banked"})
```

	id_assessment	id_student	date_submitted	score
210	1752	588775	20	85
211	1752	591774	19	88
212	1752	603861	19	61
213	1752	606143	19	78
214	1752	704156	18	67
215	1752	705379	19	81
216	1752	721259	22	null
217	1752	749412	16	83
218	1752	760729	18	80
219	1752	905042	17	64
220	1752	949618	18	63
221	1752	958987	18	78
222	1752	968578	27	62
223	1752	969076	18	82
224	1752	971027	19	76
225	1752	978739	18	59
226	1752	1035023	19	73
227	1752	1105478	19	90
228	1752	1352868	19	66
229	1752	1401935	26	45
230	1752	1402638	21	77
231	1752	1414443	50	58
232	1752	1465690	19	85
233	1752	1472925	30	61
234	1752	1488959	20	67
235	1752	1518989	19	78
236	1752	1618738	19	84

Query Settings

PROPERTIES

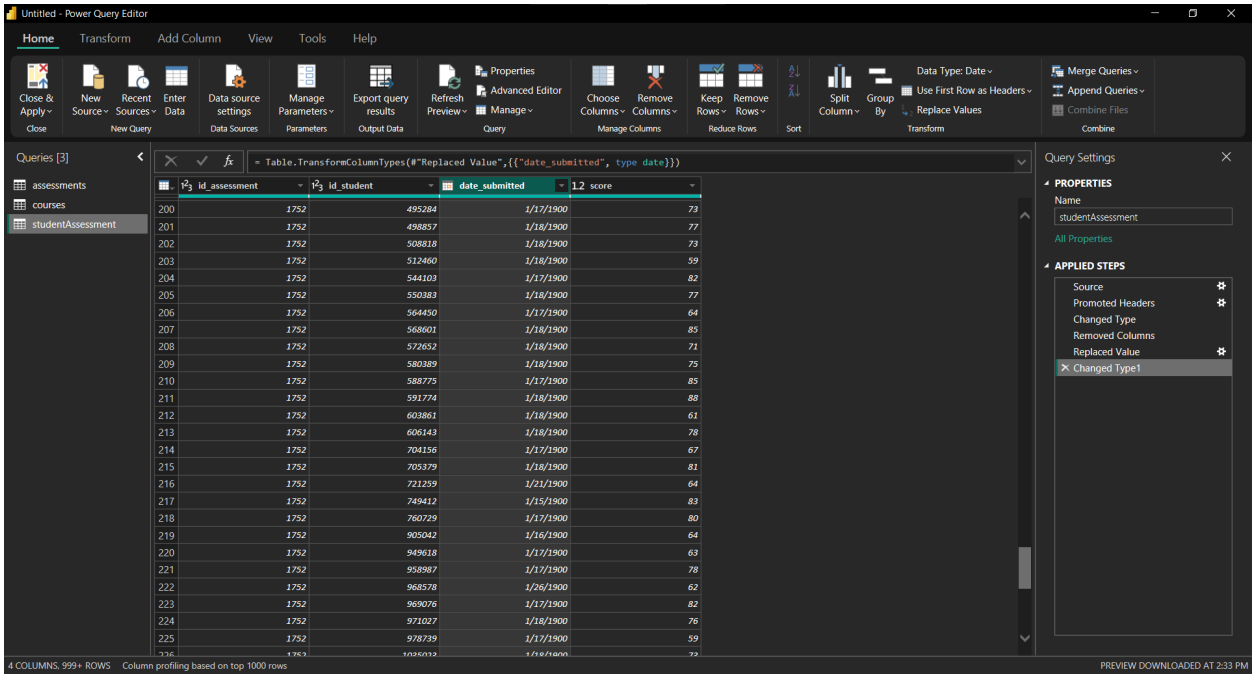
Name: studentAssessment

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Removed Columns

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:25 PM



2. Changing data type

Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type: Whole Number | Use First Row as Headers | Replace Values | Merge Queries | Append Queries | Combine Files

Queries [3]
- assessments
- courses
- studentAssessment

= Table.ReplaceValue(#"Removed Columns",null,64,Replacer.ReplaceValue,{"score"})

	id_assessment	id_student	date_submitted	score
1	1752	11391	18	78
2	1752	28400	22	70
3	1752	31604	17	72
4	1752	32885	26	69
5	1752	38053	19	79
6	1752	45462	20	70
7	1752	45642	18	72
8	1752	52130	19	72
9	1752	53025	9	71
10	1752	57506	18	68
11	1752	58873	19	73
12	1752	59185	18	67
13	1752	62155	17	73
14	1752	63400	19	83
15	1752	65002	17	66
16	1752	70464	19	59
17	1752	71361	19	82
18	1752	74372	22	60
19	1752	75091	18	67
20	1752	77367	18	73
21	1752	91265	21	75
22	1752	94961	17	74
23	1752	98094	18	62
24	1752	100893	17	63
25	1752	101781	16	84
26	1752	102806	19	80
27	1752	102952	19	76

Query Settings

PROPERTIES

Name: studentAssessment

All Properties

APPLIED STEPS
- Source
- Promoted Headers
- Changed Type
- Removed Columns
- Replaced Value

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:28 PM

Untitled - Power Query Editor

Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type: Date | Use First Row as Headers | Replace Values | Merge Queries | Append Queries | Combine Files

Queries [3]
- assessments
- courses
- studentAssessment

= Table.TransformColumnTypes(#"Replaced Value",{{"date_submitted", type date}})

	id_assessment	id_student	date_submitted	score
1	1752	11391	1/17/1900	78
2	1752	28400	1/21/1900	70
3	1752	31604	1/16/1900	72
4	1752	32885	1/25/1900	69
5	1752	38053	1/18/1900	79
6	1752	45462	1/19/1900	70
7	1752	45642	1/17/1900	72
8	1752	52130	1/18/1900	72
9	1752	53025	1/8/1900	71
10	1752	57506	1/17/1900	68
11	1752	58873	1/18/1900	73
12	1752	59185	1/17/1900	67
13	1752	62155	1/16/1900	73
14	1752	63400	1/18/1900	83
15	1752	65002	1/16/1900	66
16	1752	70464	1/18/1900	59
17	1752	71361	1/18/1900	82
18	1752	74372	1/21/1900	60
19	1752	75091	1/17/1900	67
20	1752	77367	1/17/1900	73
21	1752	91265	1/20/1900	75
22	1752	94961	1/16/1900	74
23	1752	98094	1/17/1900	62
24	1752	100893	1/16/1900	63
25	1752	101781	1/15/1900	84
26	1752	102806	1/18/1900	80
27	1752	102952	1/18/1900	76

Query Settings

PROPERTIES

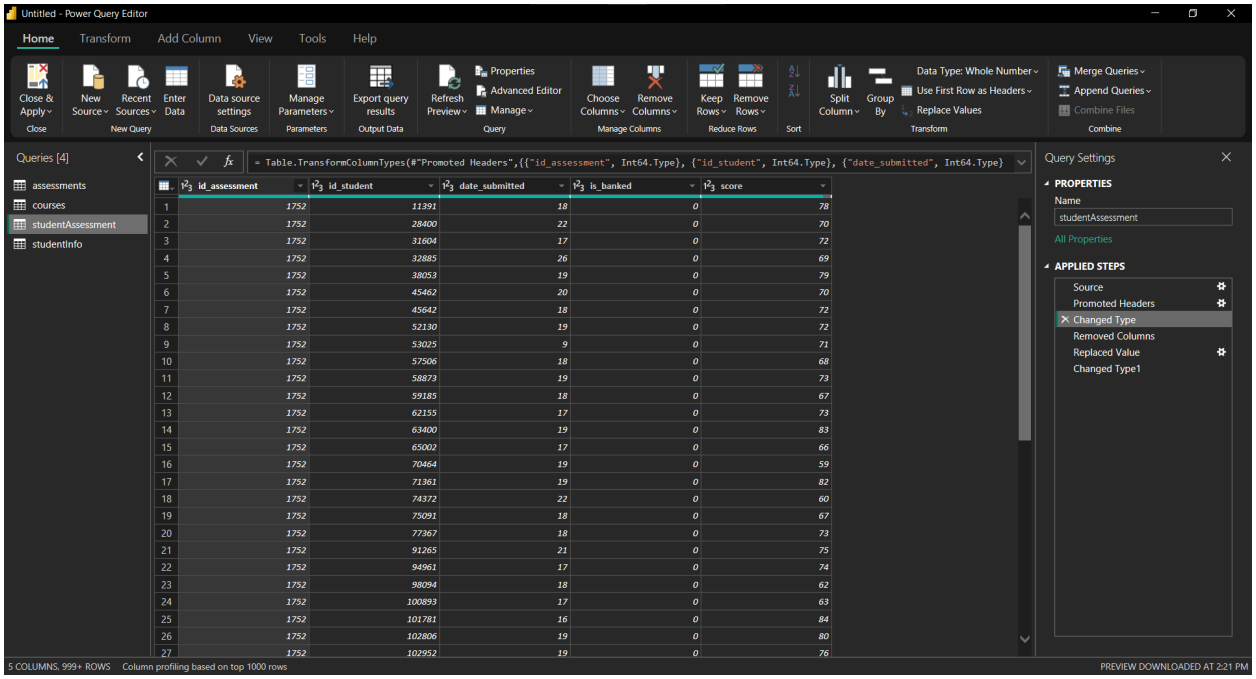
Name: studentAssessment

All Properties

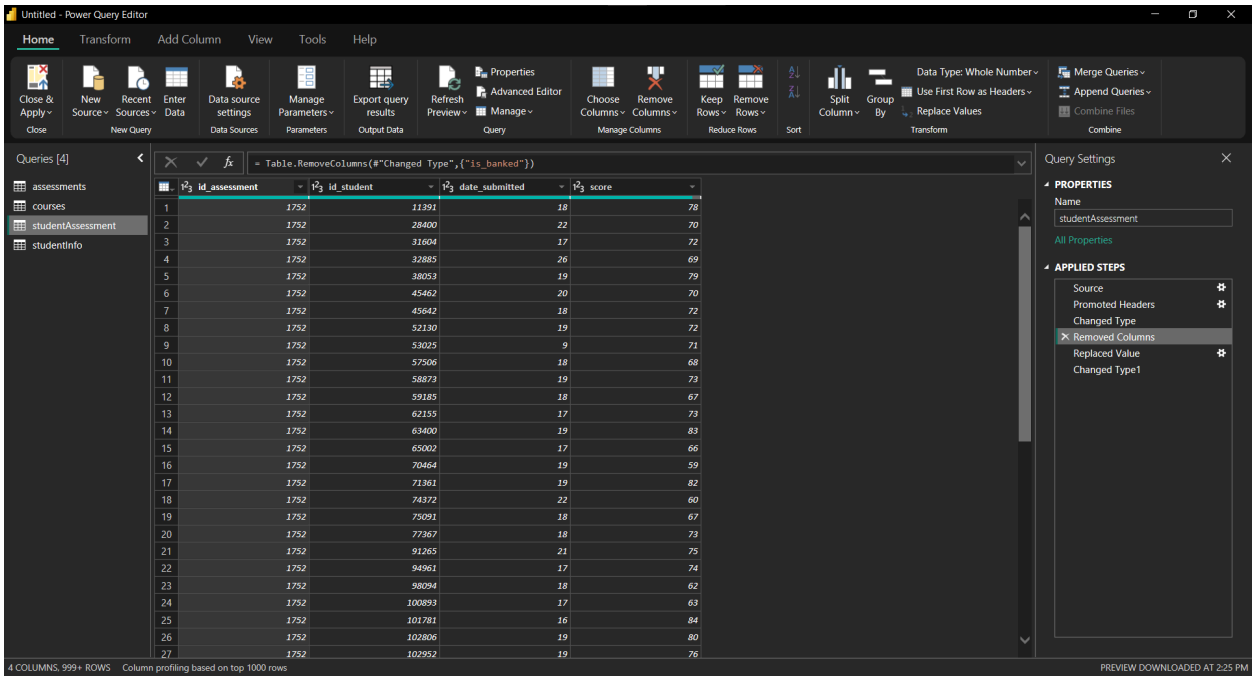
APPLIED STEPS
- Source
- Promoted Headers
- Changed Type
- Removed Columns
- Replaced Value
- Changed Type1

4 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 2:33 PM

3. Remove columns



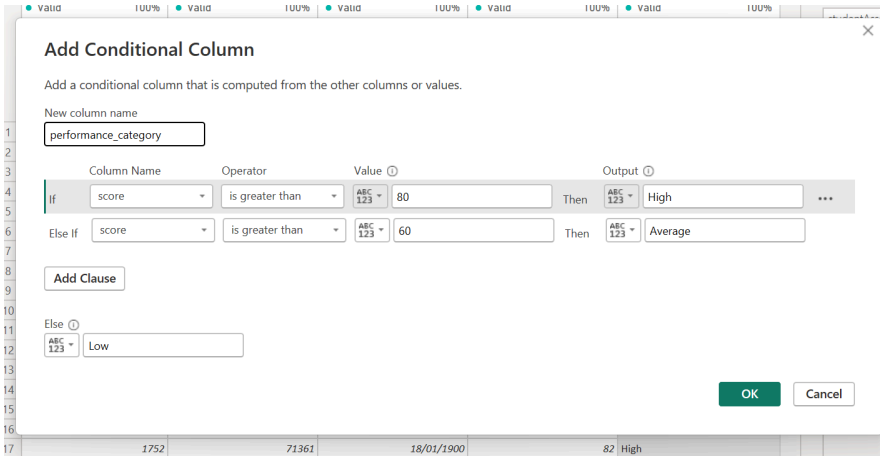
	id_assessment	id_student	date_submitted	is_banked	score
1	1752	11391	18	0	78
2	1752	28400	22	0	70
3	1752	31604	17	0	72
4	1752	32885	26	0	69
5	1752	38053	19	0	79
6	1752	45462	20	0	70
7	1752	45642	18	0	72
8	1752	52130	19	0	72
9	1752	53025	9	0	71
10	1752	57506	18	0	68
11	1752	58873	19	0	73
12	1752	59185	18	0	67
13	1752	62155	17	0	73
14	1752	63400	19	0	83
15	1752	65002	17	0	66
16	1752	70464	19	0	59
17	1752	71361	19	0	82
18	1752	74372	22	0	60
19	1752	75091	18	0	67
20	1752	77367	18	0	73
21	1752	91265	21	0	75
22	1752	94961	17	0	74
23	1752	98094	18	0	62
24	1752	100893	17	0	63
25	1752	101781	16	0	84
26	1752	102806	19	0	80
27	1752	102952	19	0	76



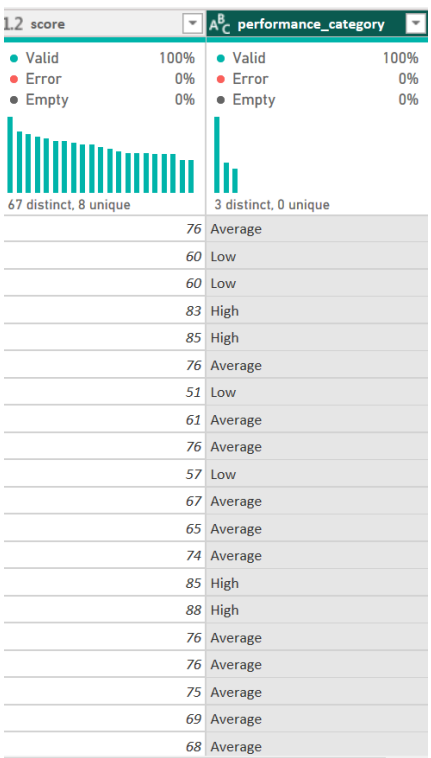
	id_assessment	id_student	date_submitted	score
1	1752	11391	18	78
2	1752	28400	22	70
3	1752	31604	17	72
4	1752	32885	26	69
5	1752	38053	19	79
6	1752	45462	20	70
7	1752	45642	18	72
8	1752	52130	19	72
9	1752	53025	9	71
10	1752	57506	18	68
11	1752	58873	19	73
12	1752	59185	18	67
13	1752	62155	17	73
14	1752	63400	19	83
15	1752	65002	17	66
16	1752	70464	19	59
17	1752	71361	19	82
18	1752	74372	22	60
19	1752	75091	18	67
20	1752	77367	18	73
21	1752	91265	21	75
22	1752	94961	17	74
23	1752	98094	18	62
24	1752	100893	17	63
25	1752	101781	16	84
26	1752	102806	19	80
27	1752	102952	19	76

4. Conditional Column – studentAssessment

Created a performance category based on the assessment score (High / Medium / Low).



The Result:



score	performance_category
76	Average
60	Low
60	Low
83	High
85	High
76	Average
51	Low
61	Average
76	Average
57	Low
67	Average
65	Average
74	Average
85	High
88	High
76	Average
76	Average
75	Average
69	Average
68	Average

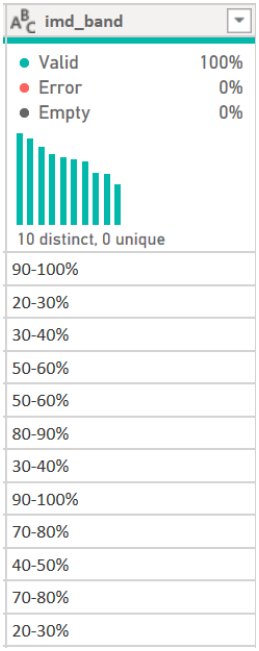
5. Split imd_band into lower and upper

imd_band represents the socio-economic Index of Multiple Deprivation (IMD) for the student's home area.
It is expressed as a percentage range (e.g., *"0–10%", "20–30%", "80–90%"*), where:

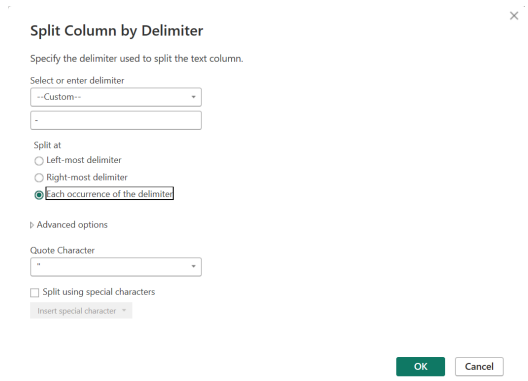
- **Lower band values** = more deprived areas

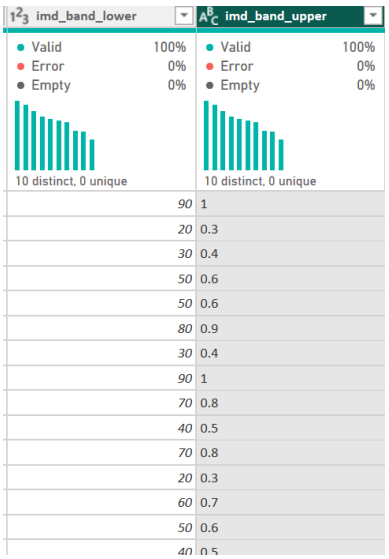
- **Higher band values** = more advantaged areas

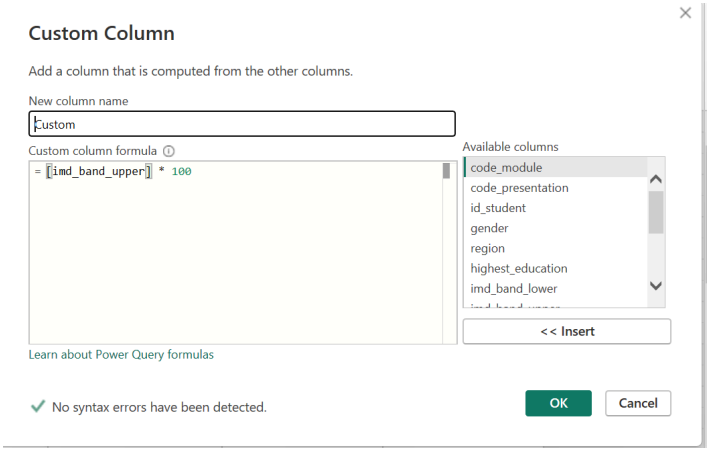
Before:



Process:







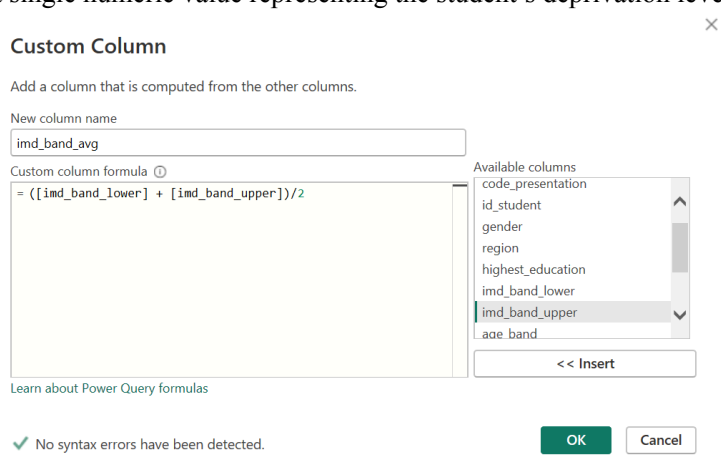
Final:

imd_band_lower	imd_band_upper
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
10 distinct, 0 unique	
90	100
20	30
30	40
50	60
50	60
80	90
30	40
90	100
70	80
40	50
70	80
20	30
60	70
50	60

6. Custom column – imd_band_avg

Created a new column to calculate the average of the IMD range.
This gives a single numeric value representing the student's deprivation level.

Process:



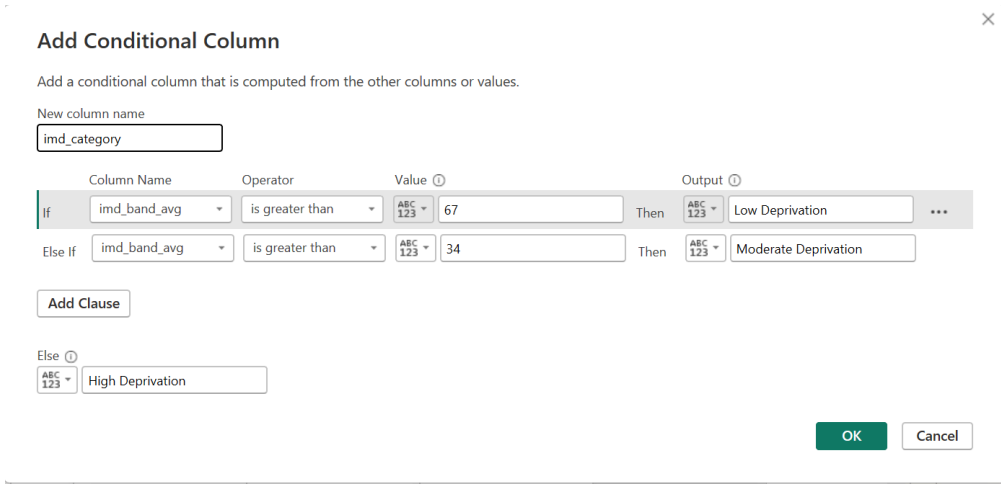
The result:

imd_band_avg
Valid 100%
Error 0%
Empty 0%
95
25
35
55
55
85
35

7. **Conditional column – Imd_category**

Classified students based on their imd_band_avg into:

- High Deprivation
- Moderate Deprivation
- Low Deprivation



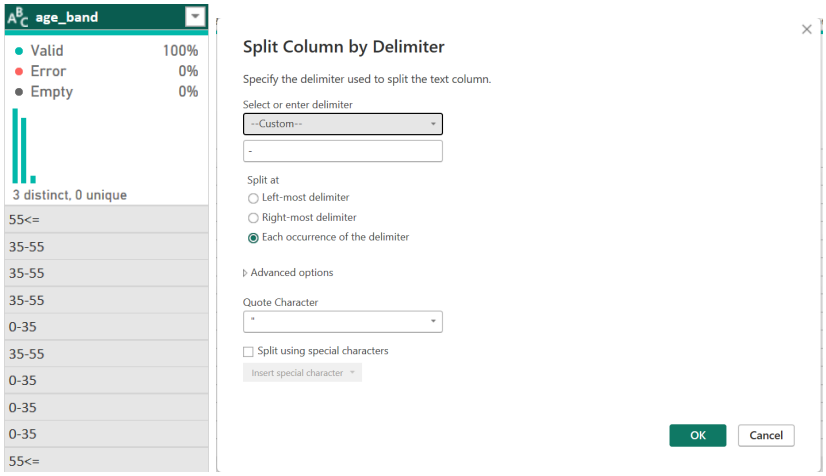
The result:

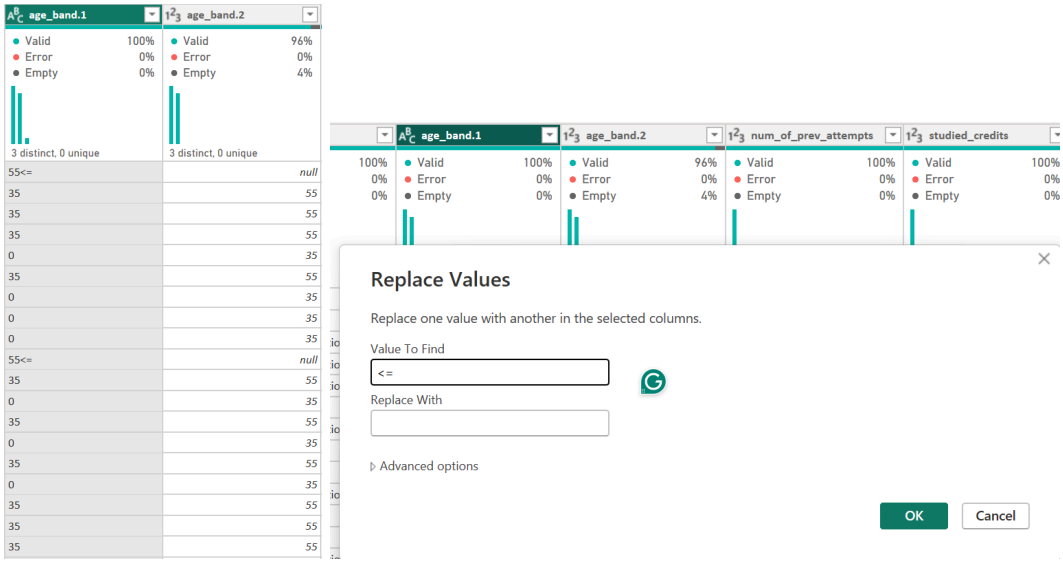
imd_band_avg	imd_category
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
95	Low Deprivation
25	High Deprivation
35	Moderate Deprivation
55	Moderate Deprivation
55	Moderate Deprivation
85	Low Deprivation
35	Moderate Deprivation
95	Low Deprivation
75	Low Deprivation

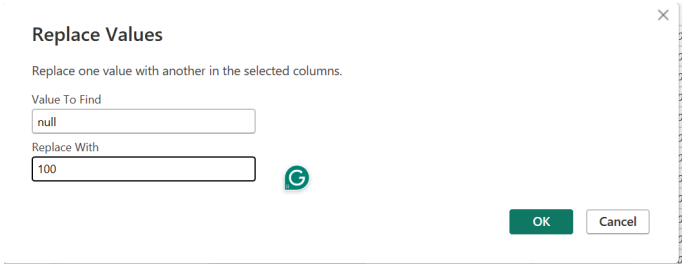
8. **Split the age_band**

The original age bands (e.g., *"0-35", "35-55", "55<="*) were split into numeric minimum and maximum age values.

Before


age_band
Valid 100%
Error 0%
Empty 0%
3 distinct, 0 unique
55<=
35-55
35-55
35-55
0-35
35-55
0-35
0-35
0-35
55<=
Split Column by Delimiter
Specify the delimiter used to split the text column.
Select or enter delimiter
--Custom--
-
Split at
Left-most delimiter
Right-most delimiter
Each occurrence of the delimiter
Advanced options
Quote Character
"
Split using special characters
Insert special character
OK
Cancel


age_band.1
age_band.2
Valid 100% Valid 96%
Error 0% Error 0%
Empty 0% Empty 4%
3 distinct, 0 unique
3 distinct, 0 unique
num_of_prev_attempts
studied_credits
Replace Values
Replace one value with another in the selected columns.
Value To Find
<=
Replace With
Advanced options
OK
Cancel


Replace Values
Replace one value with another in the selected columns.
Value To Find
null
Replace With
100
OK
Cancel

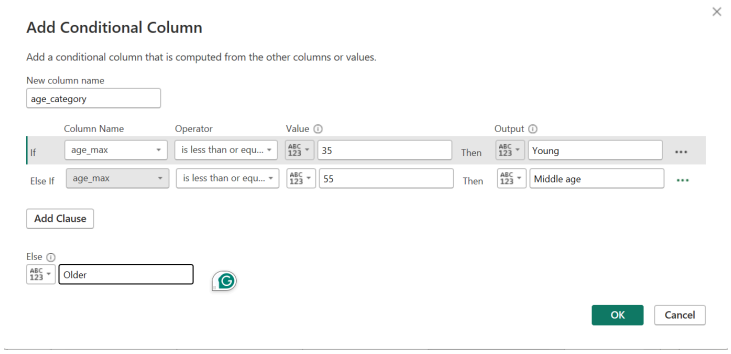
Final:

age_min	age_max
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
3 distinct, 0 unique	3 distinct, 0 unique
55	100
35	55
35	55
35	55
0	35
35	55
0	35
0	35
0	35
55	100
35	55
0	35
35	55
0	35
35	55
0	35
35	55
35	55
35	55
35	55

9. **Conditional column – age_category**

   Created age groups:

- Young (≤35)
- Middle Age (36–55)
- Older (55+)



Final:

age_min	age_max	age_category
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
3 distinct, 0 unique	3 distinct, 0 unique	
55	100	Older
35	55	Middle age
35	55	Middle age
35	55	Middle age
0	35	Young
35	55	Middle age
0	35	Young
0	35	Young
0	35	Young
55	100	Older
35	55	Middle age
0	35	Young
35	55	Middle age
0	35	Young
35	55	Middle age
0	35	Young
35	55	Middle age
35	55	Middle age
35	55	Middle age
35	55	Middle age

10. Renaming and cleaning the date columns

a.) studentVle - Interaction dtate

The "date" column actually represents:

"The date of the student's interaction measured as the number of days since the start of the module-presentation."

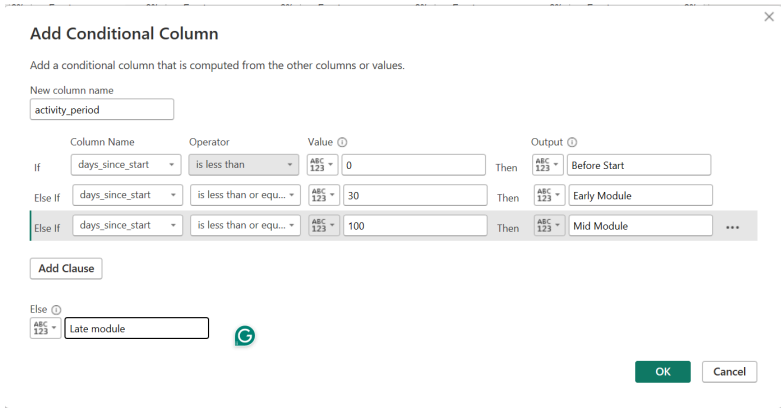
Before:

date
Valid 100%
Error 0%
Empty 0%
39 distinct, 6 unique
-10
-10
-10
-10
-10
-10
-10
-10
-10
-10
-10
-10
-10

After:

days_since_start
Valid 100%
Error 0%
Empty 0%
39 distinct, 6 unique
-10
-10
-10
-10
-10
-10
-10
-10
-10
-10

Creating a conditional column to analyse when students engage the most



Final:

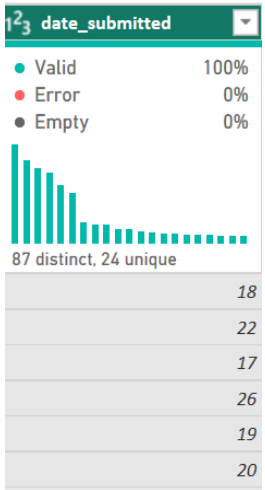
days_since_start	activity_period
Valid 100%	Valid 100%
Error 0%	Error 0%
Empty 0%	Empty 0%
39 distinct, 6 unique	
-4	Before Start
-4	Before Start
-4	Before Start
-4	Before Start
-3	Before Start
-3	Before Start
-3	Before Start
-2	Before Start
-2	Before Start
-1	Before Start
2	Early Module
2	Early Module
2	Early Module
2	Early Module
2	Early Module
2	Early Module
3	Early Module
3	Early Module
5	Early Module

b.) studentAssessment

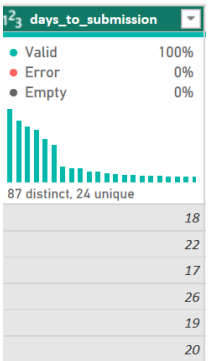
According to the dataset:

"The date of student submission is measured as the number of days since the start of the module-presentation.

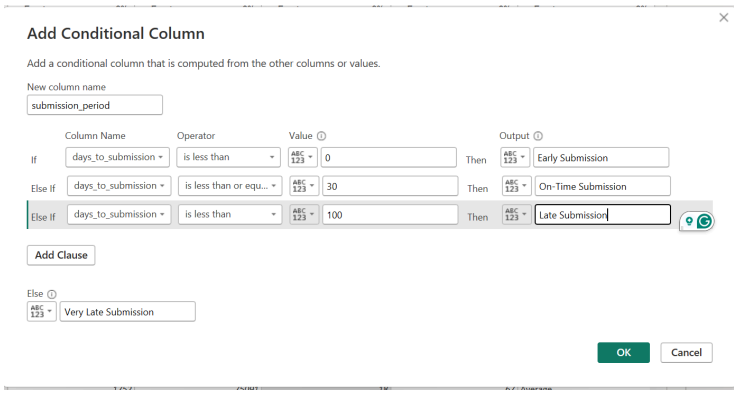
Before:



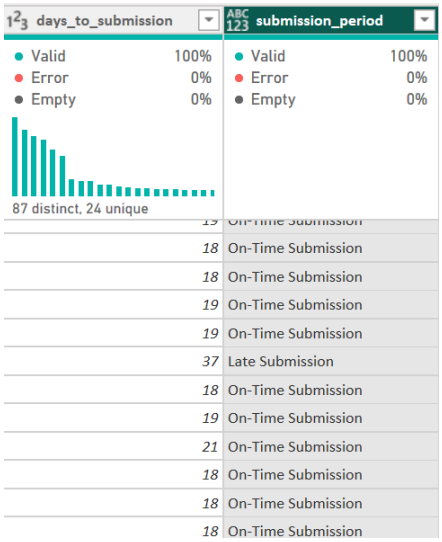
After:



Created a conditional column to analyse submission behaviour (Early / On-time / Late / Very Late).



After:

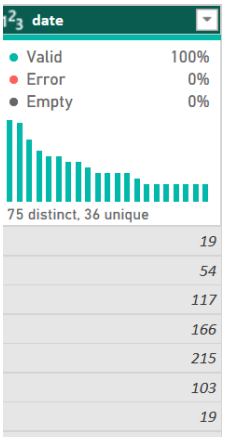


c.) Assessments Table — Final Submission Date

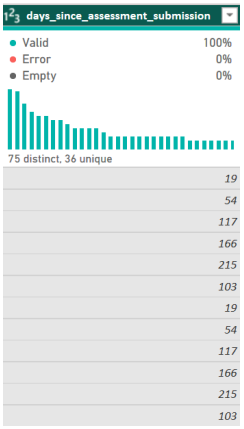
The “date” column represents:

“Information about the final submission date of the assessment, calculated as the number of days since the start of the module-presentation.”

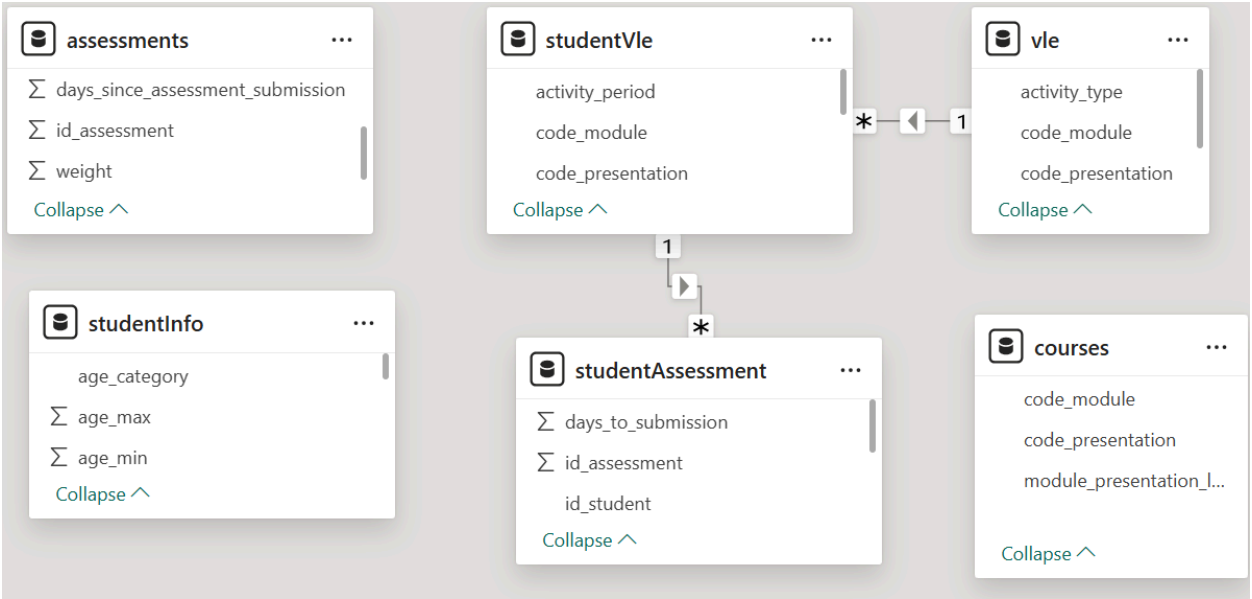
Before:



After:

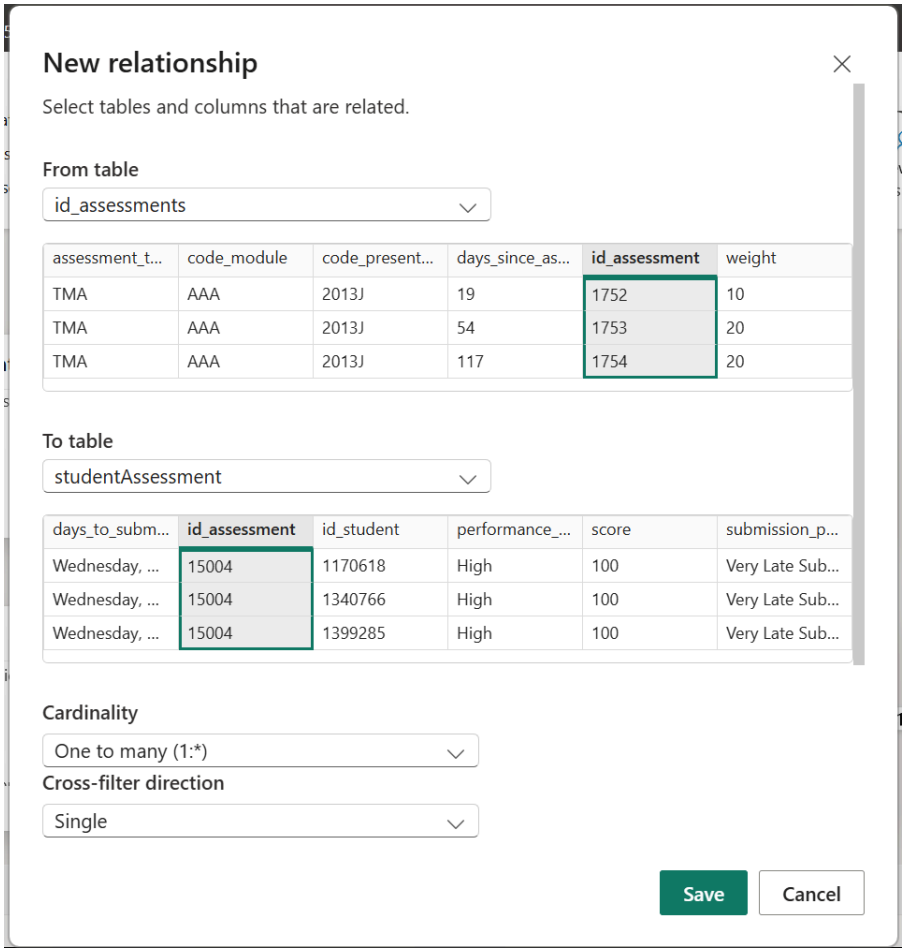


Model View:



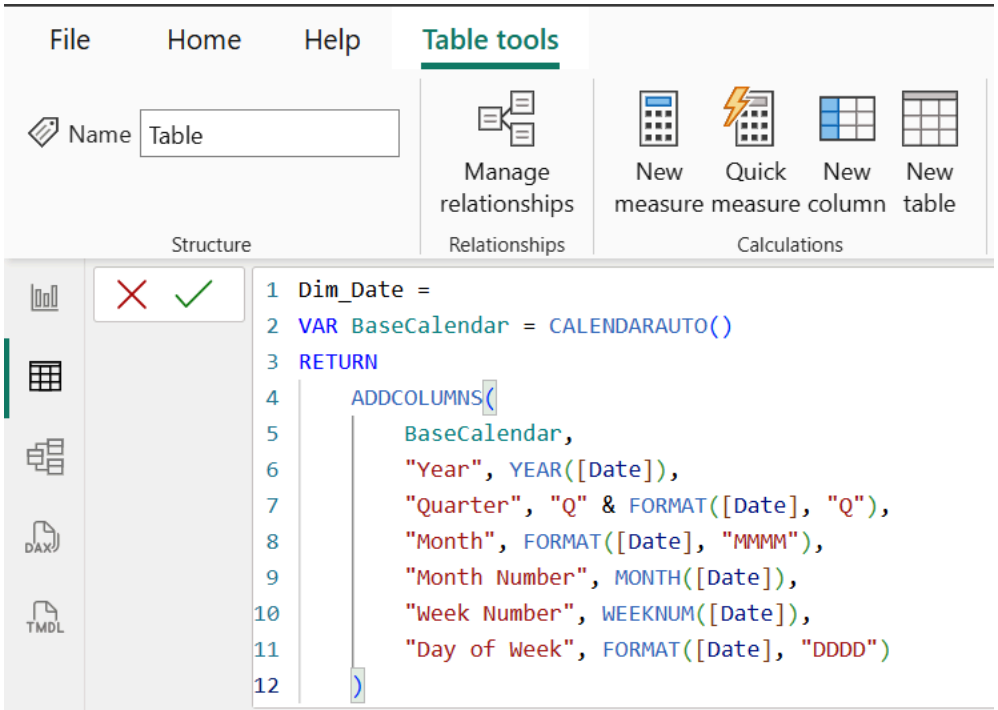
Before

- This shows the initial model before any changes were made.
- At this stage, the tables are not yet well connected or optimized for analysis.

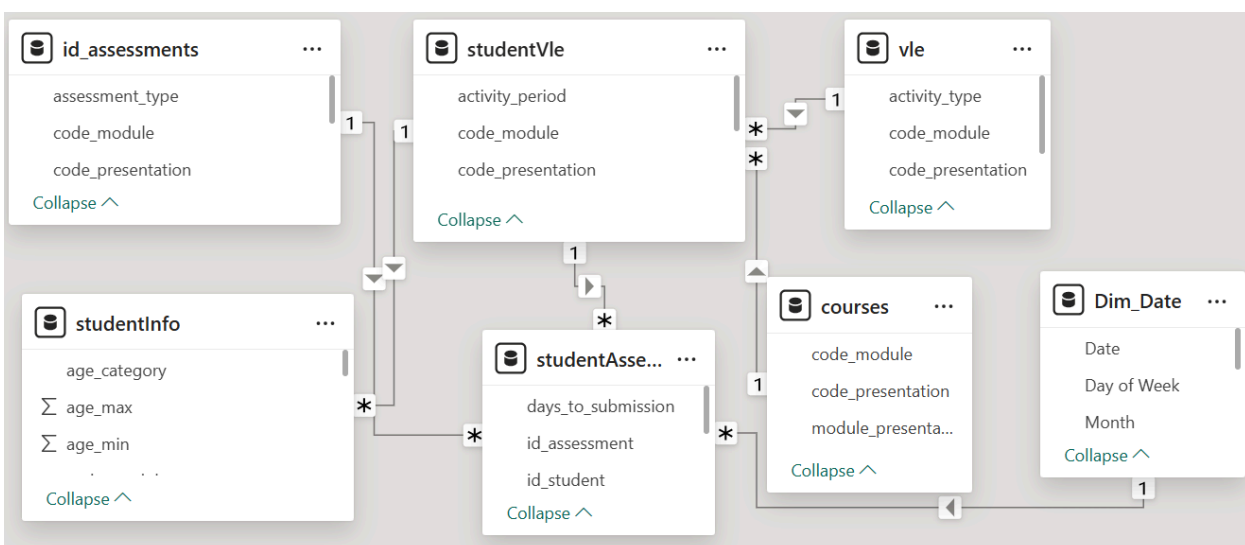


Creating new relationships

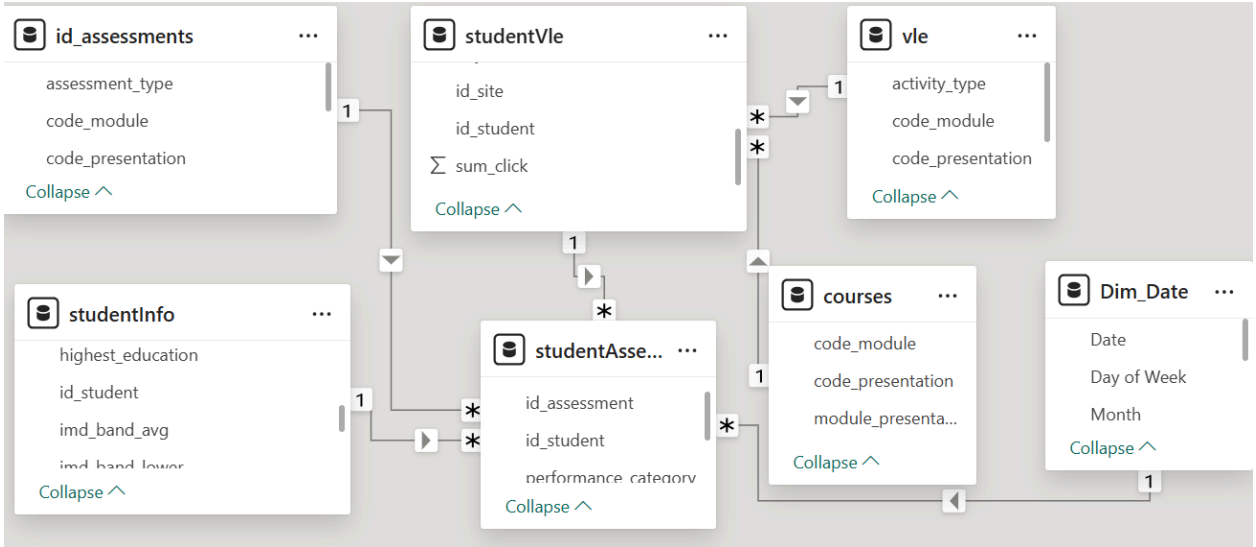
- Relationships were added to link the tables together.
- This makes the data easier to work with and ensures more accurate analysis.



Calculated DAX Table
- A new table was created using DAX formulas.
- This helps organize the data better and supports additional calculations.



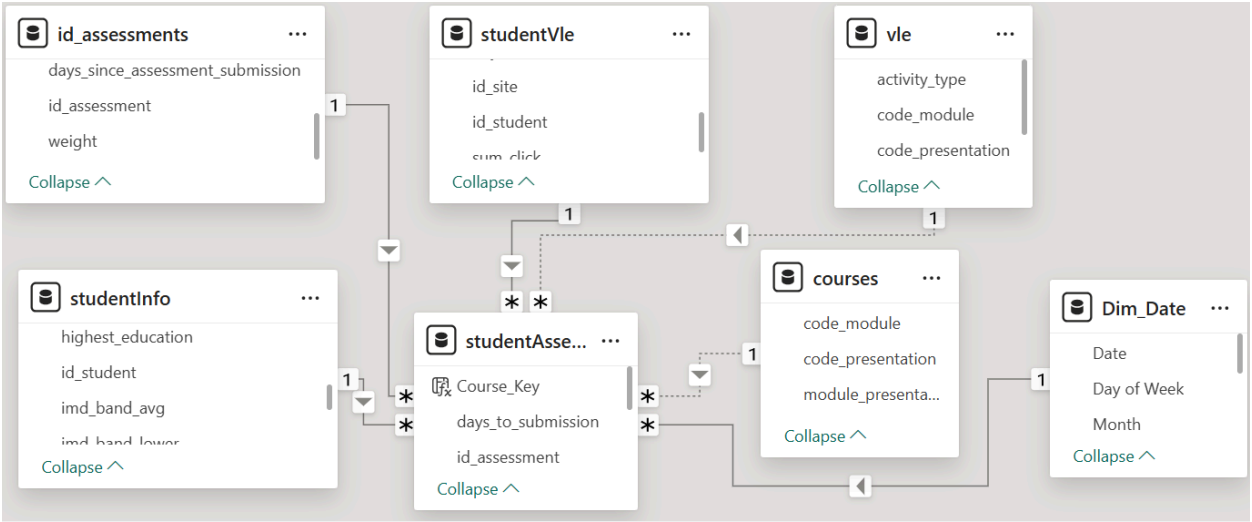
Model View including the newly created DAX Table Dim_Date
- The Dim_Date table was added to handle date-related analysis.
- This makes it easier to analyze trends over time.




```
Course_Key = RELATED('id_assessments'[code_module]) & "-" & RELATED('id_assessments'[code_presentation])
```

id_student	days_to_submission	score	performance_category	submission_period	Course_Key
1170618	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1340766	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1399285	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1405594	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1421720	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1440341	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1467014	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J
1503056	Wednesday, April 11, 1900	100	High	Very Late Submission	BBB-2013J

Creating a new column for star schema in the studentAssesment table



Final model view with the correct cardinality

- The relationships were adjusted to the correct cardinality (one-to-many).
- The model is now clean, well-structured, and ready for accurate reporting.

Dax Measures:

Basic measures

Name: Total Courses
Home table: courses
Format: Whole number
Data category: Uncategorized

```
Total Courses = DISTINCTCOUNT(courses[code_module])
```

Name: Total Students
Home table: studentInfo
Format: Whole number
Data category: Uncategorized

```
Total Students = DISTINCTCOUNT(studentInfo[id_student])
```

VLE measures

Name: Total Clicks
Home table: studentVle
Format: Whole number
Data category: Uncategorized

```
Total Clicks = SUM(studentVle[sum_click])
```

Name: Avg Clicks per Stu...
Home table: studentVle
Format: General
Data category: Uncategorized

```
Avg Clicks per Student =
DIVIDE([Total Clicks], [Total Students])
```

Assessment measures

Name: Average Score
Home table: studentAssessment
Format: General
Data category: Uncategorized

```
Average Score = AVERAGE(studentAssessment[score])
```

(Assuming pass = score ≥ 40)

Name: Pass Rate
Home table: studentAssessment
Format: General
Auto
Data category: Uncategorized
Structure | Formatting | Properties

```
Pass Rate =
DIVIDE(
    CALCULATE(COUNTROWS(studentAssessment), studentAssessment[score] >= 40),
    COUNTROWS(studentAssessment)
)
```

Student Outcomes measures

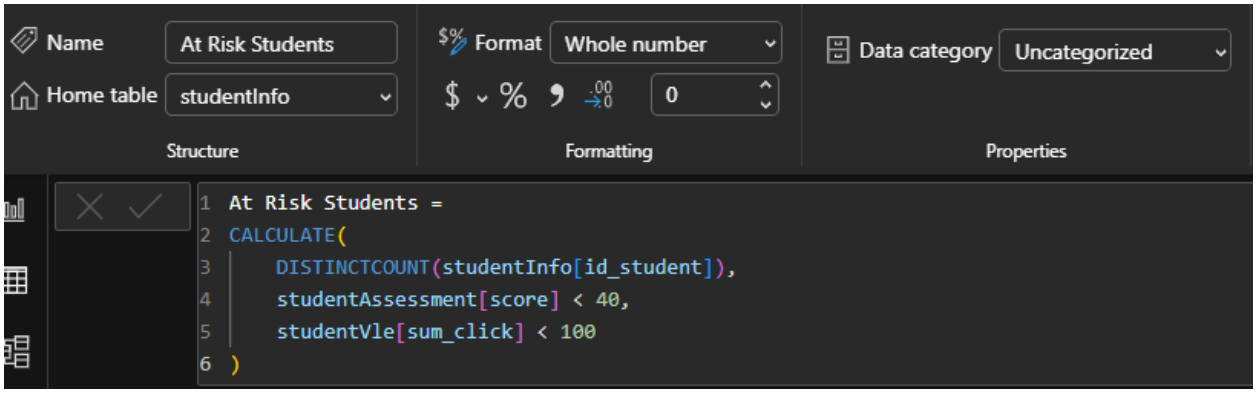
Name: Completion Rate
Home table: studentInfo
Format: General
Auto
Data category: Uncategorized
Structure | Formatting | Properties

```
Completion Rate =
DIVIDE(
    CALCULATE(COUNTROWS(studentInfo), studentInfo[final_result] = "Pass"),
    COUNTROWS(studentInfo)
)
```

Name: Dropout Rate
Home table: studentInfo
Commit
Format: General
Auto
Data category: Uncategorized
Formatting | Properties

```
Dropout Rate =
DIVIDE(
    CALCULATE(COUNTROWS(studentInfo), studentInfo[final_result] = "Withdrawn"),
    COUNTROWS(studentInfo)
)
```

Risk indicator

At-Risk Students

(low engagement + low score)